

Hybrid EMI Filter for 11kW Onboard Charger

Weihao Zhao

Stefan Mollov

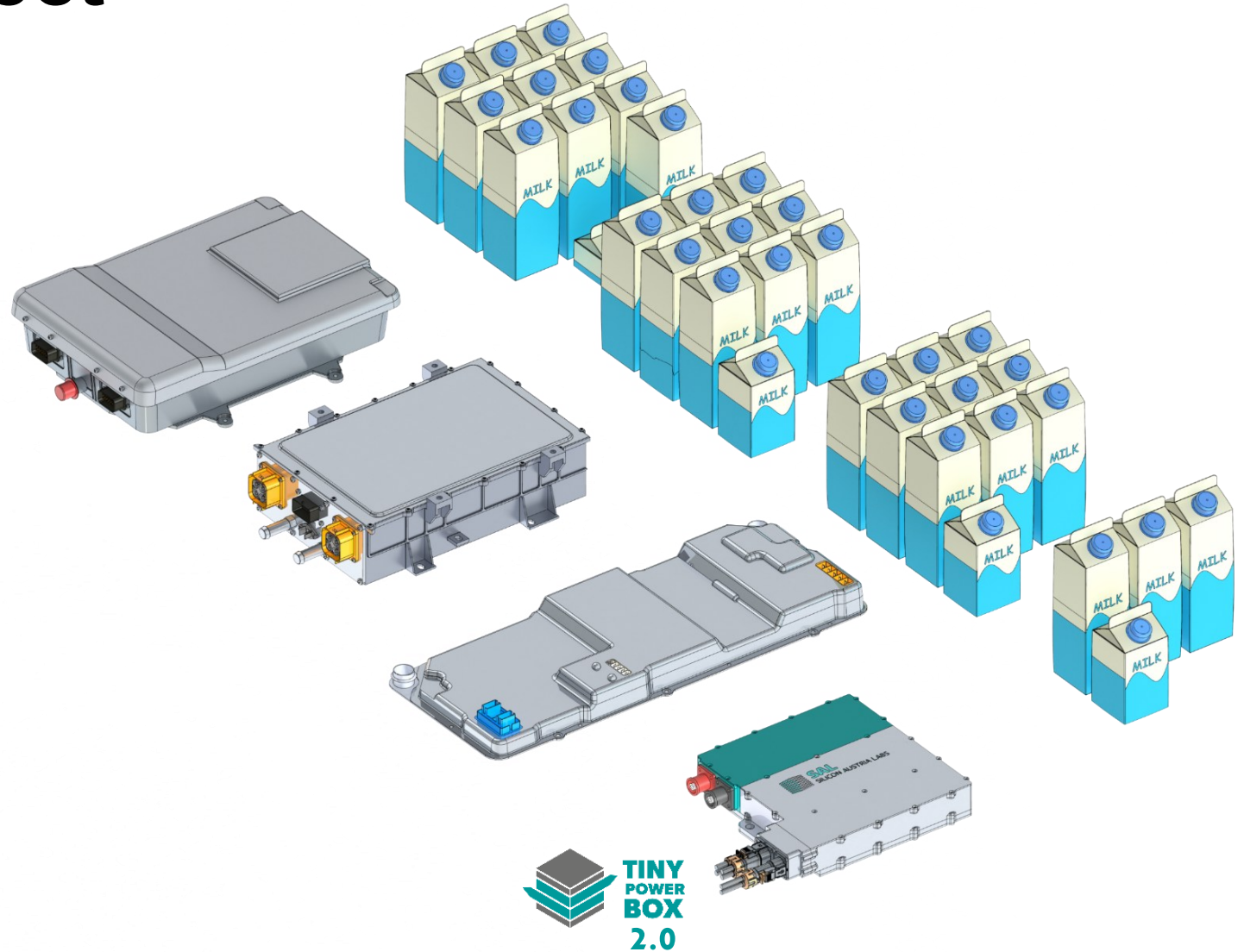
Michael Hartmann

Tiny Power Box Project

*Volume indicator

High-Power Density OBC

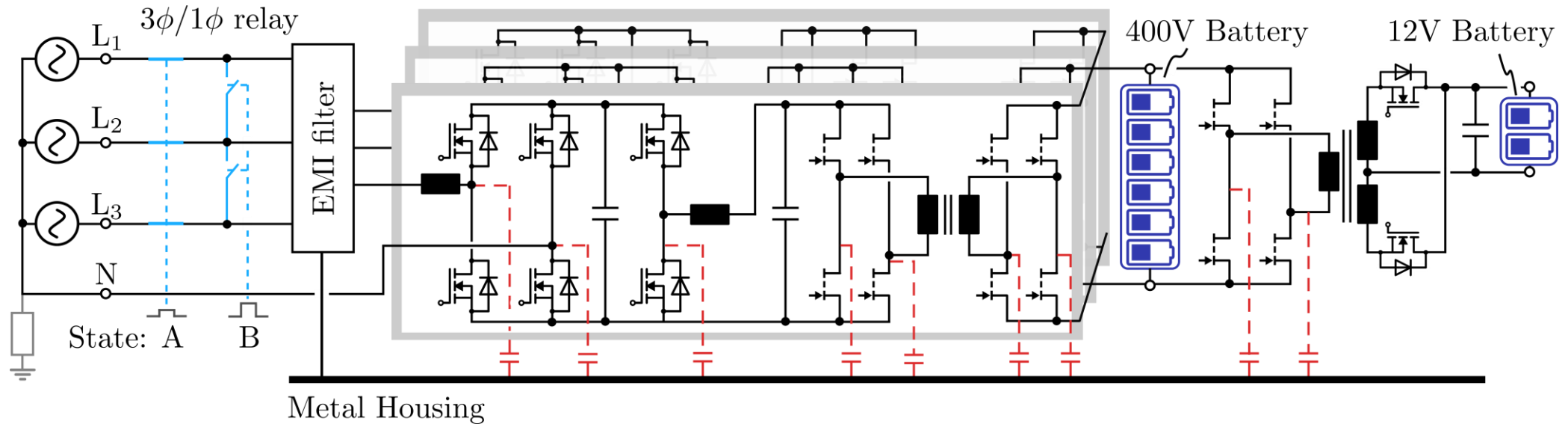
- 3 ϕ (11kW) / 1 ϕ (7.2kW)
- IEC-61851 Compliant
- Low-Profile



3- ϕ /1- ϕ Input EV Chargers

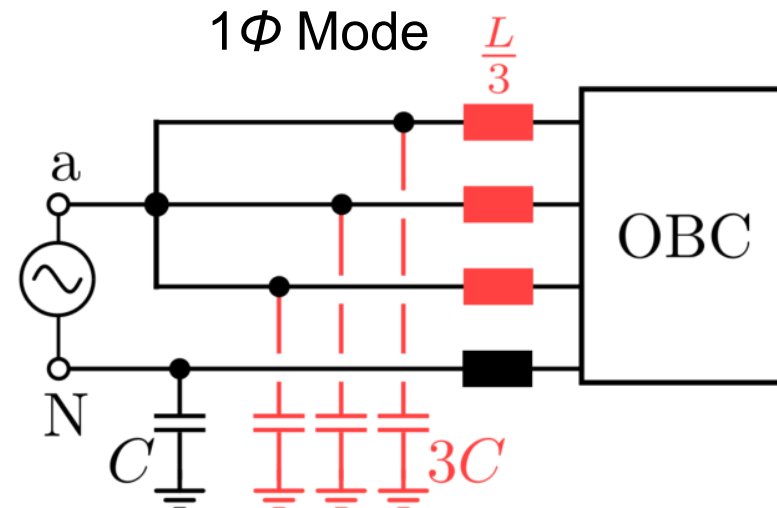
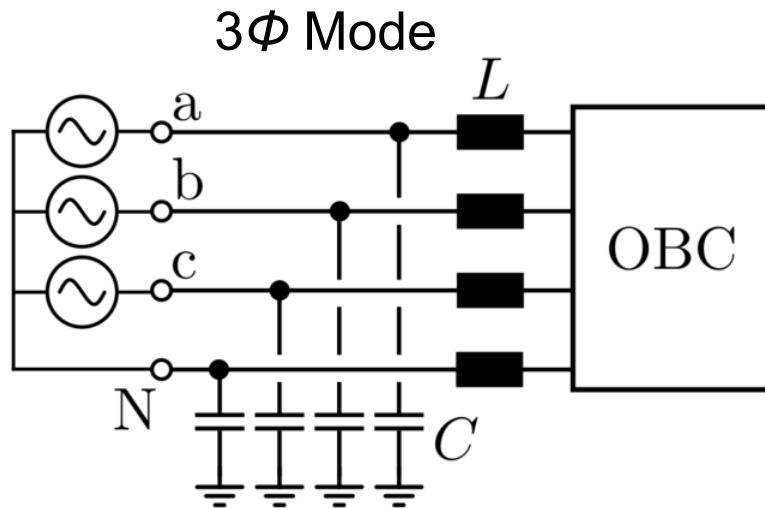
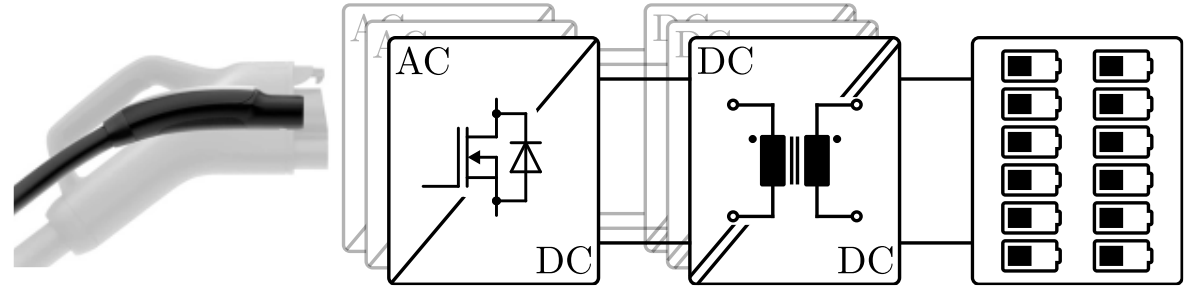
High-Power Density OBC

- 3 ϕ (11kW) / 1 ϕ (7.2kW)
- IEC-61851 Compliant
- Low-Profile



Design Challenges

- $3\phi/1\phi$ Input Compatible



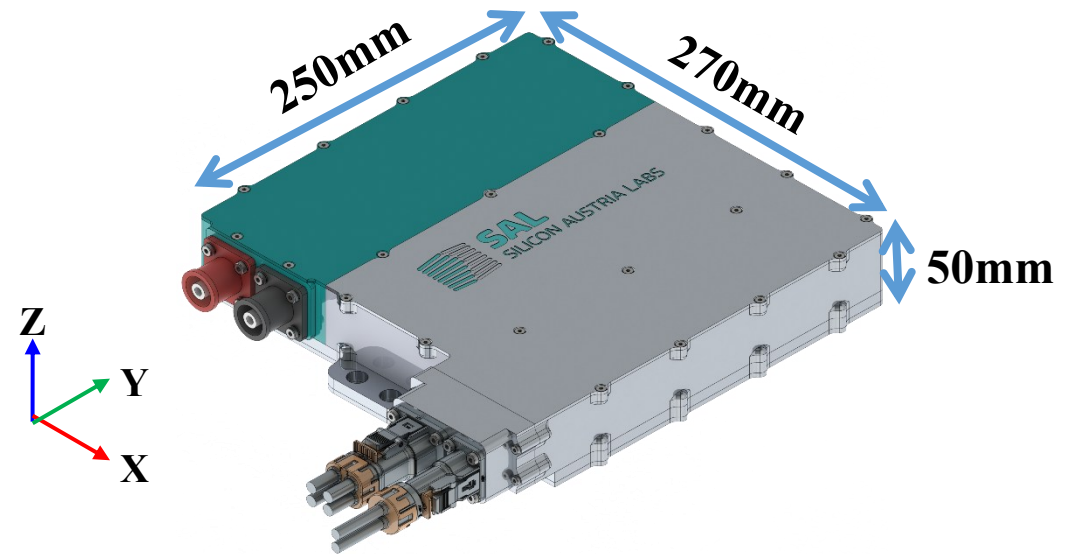
Design Challenges

- 3 ϕ /1 ϕ Input Compatible
- Low Leakage Current^[1]



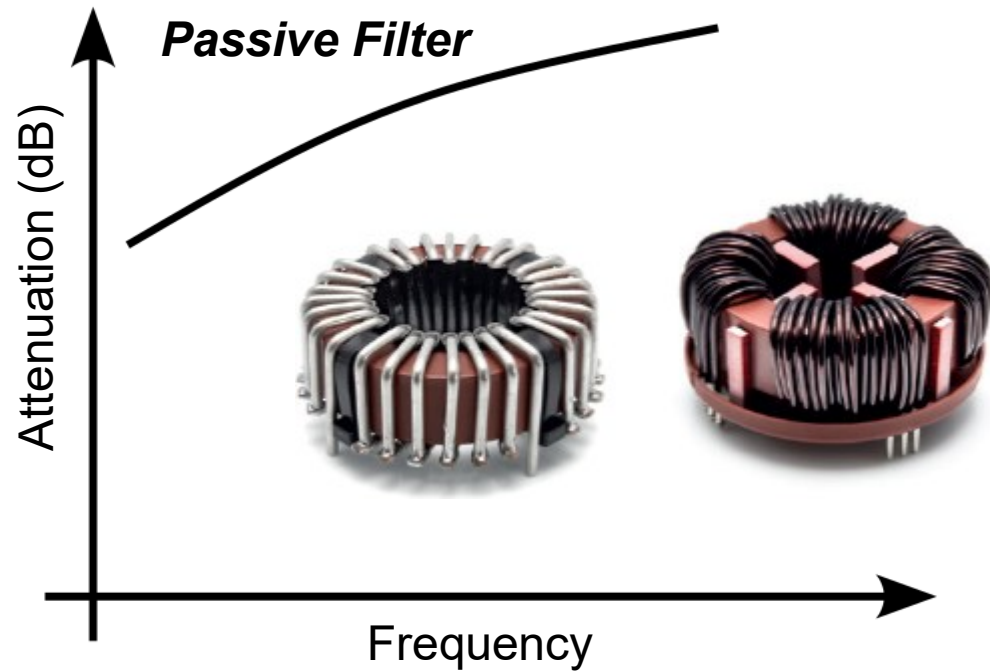
Design Challenges

- $3\phi/1\phi$ Input Compatible
- Low Leakage Current^[1]
- Strict Volume Restriction

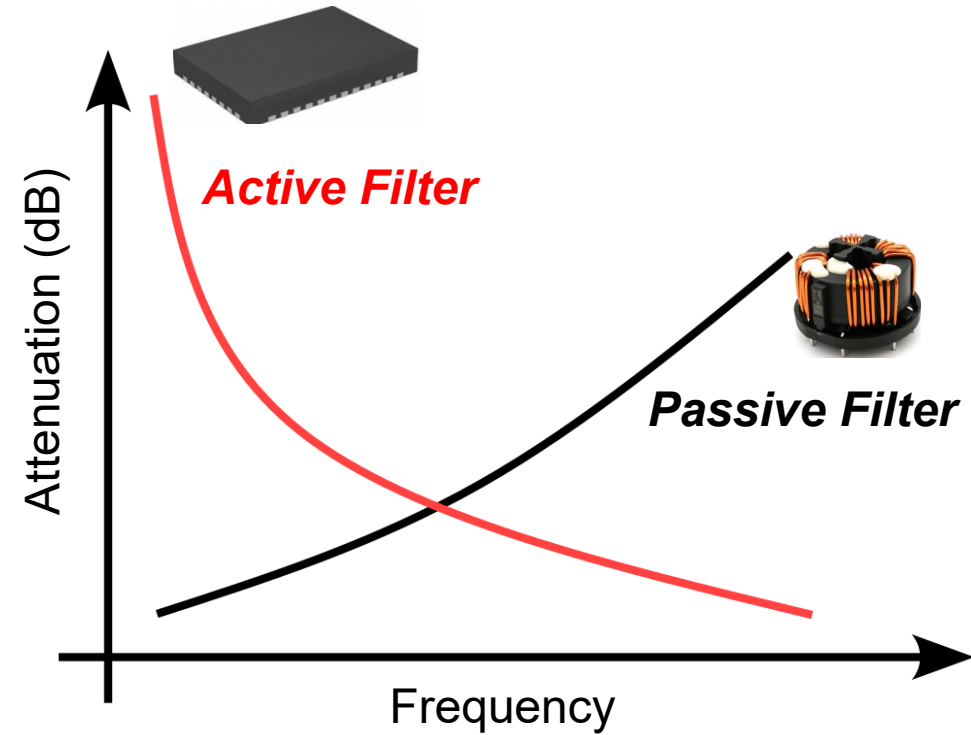


Solution

Passive EMI Filter

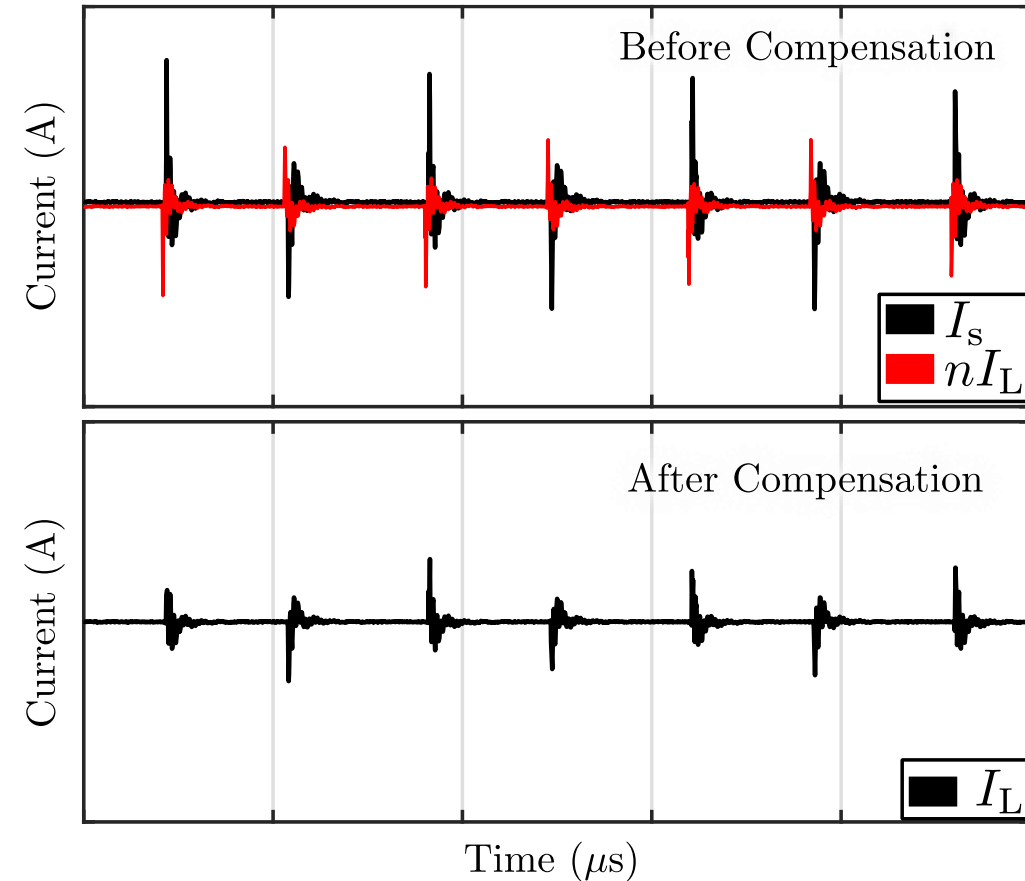
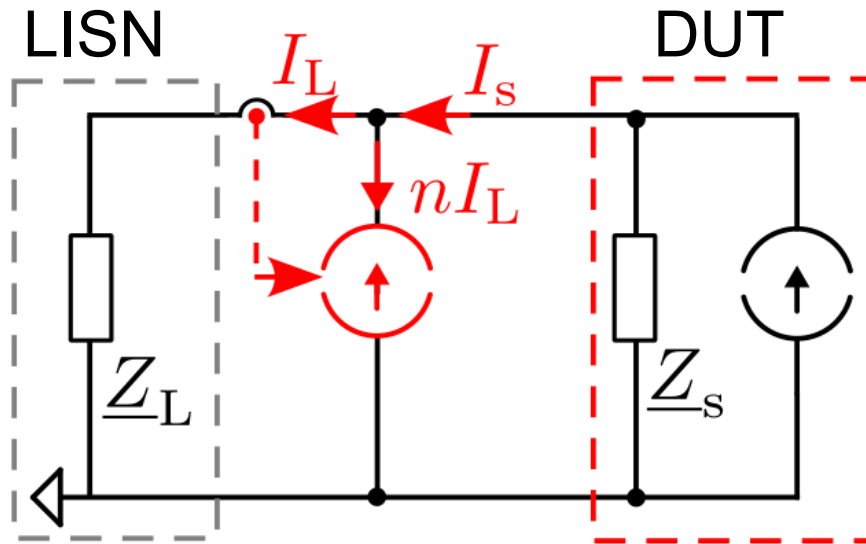


Hybrid EMI Filter



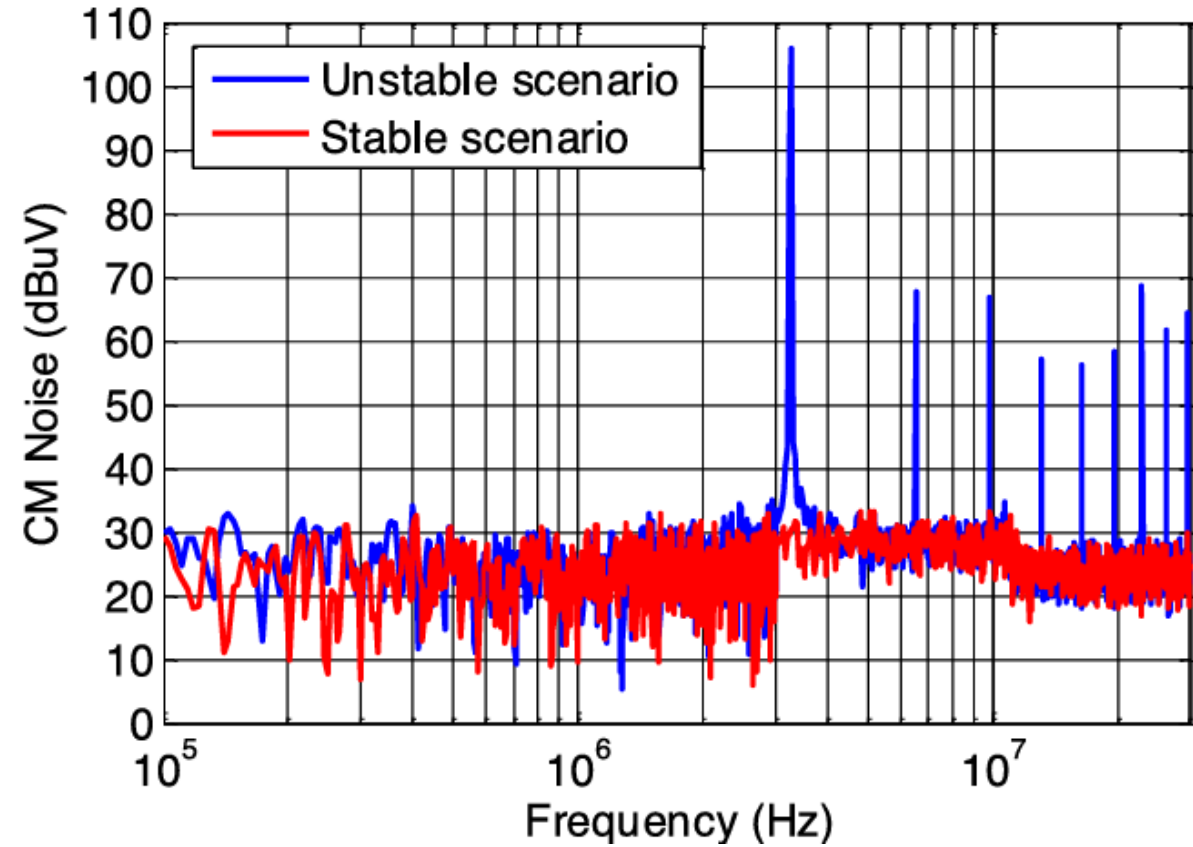
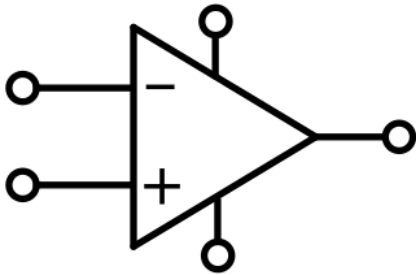
Active EMI Filter

- Measurement
- Processing
- Compensation



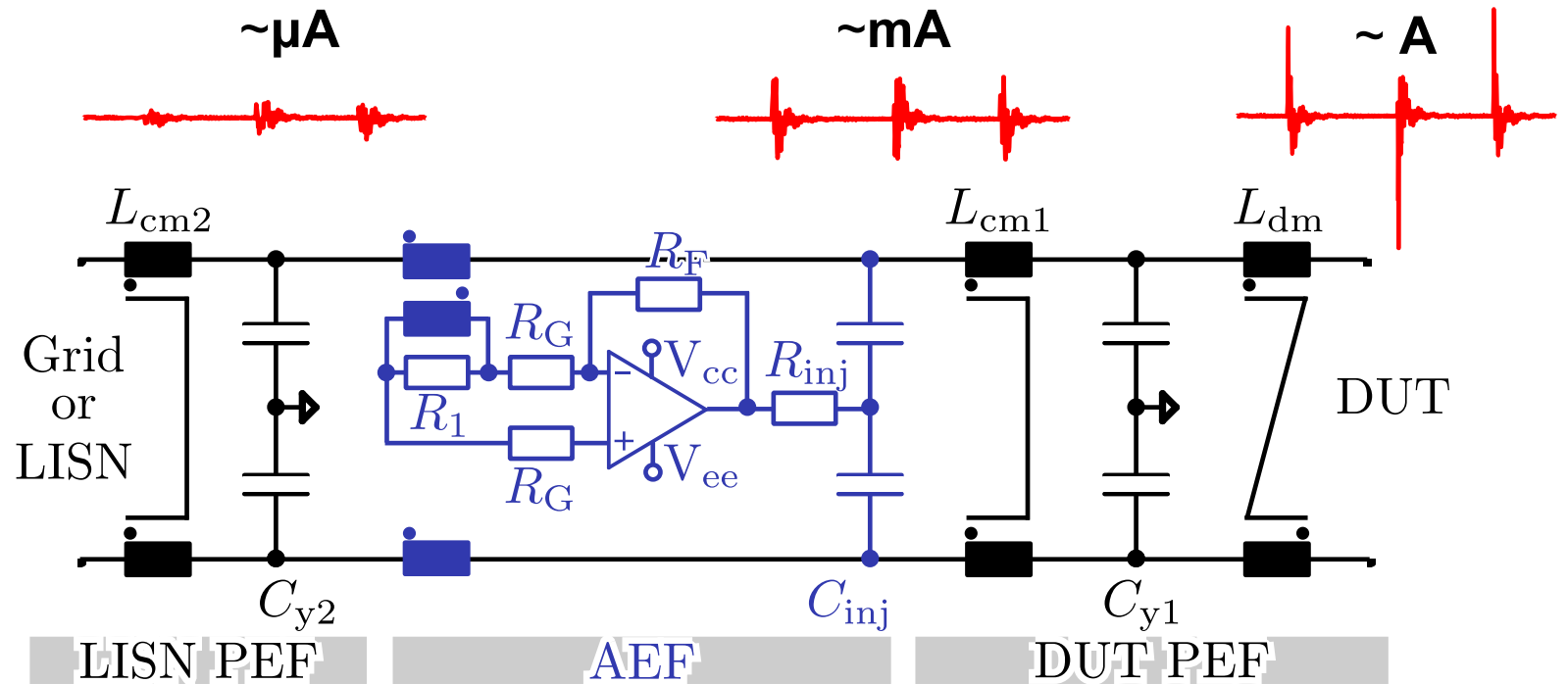
Design Consideration

- Output Capability
- Stability
- Adaptability



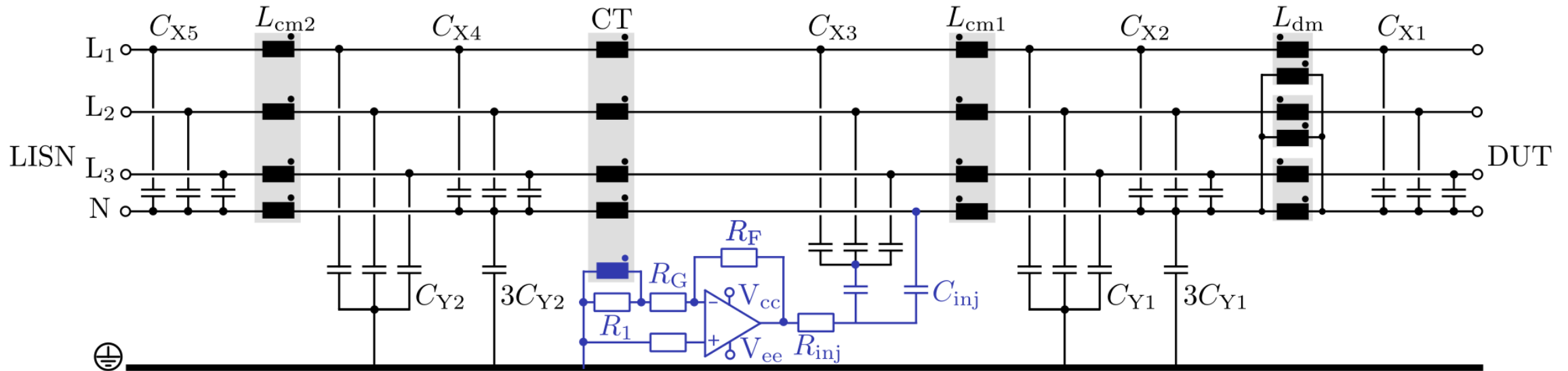
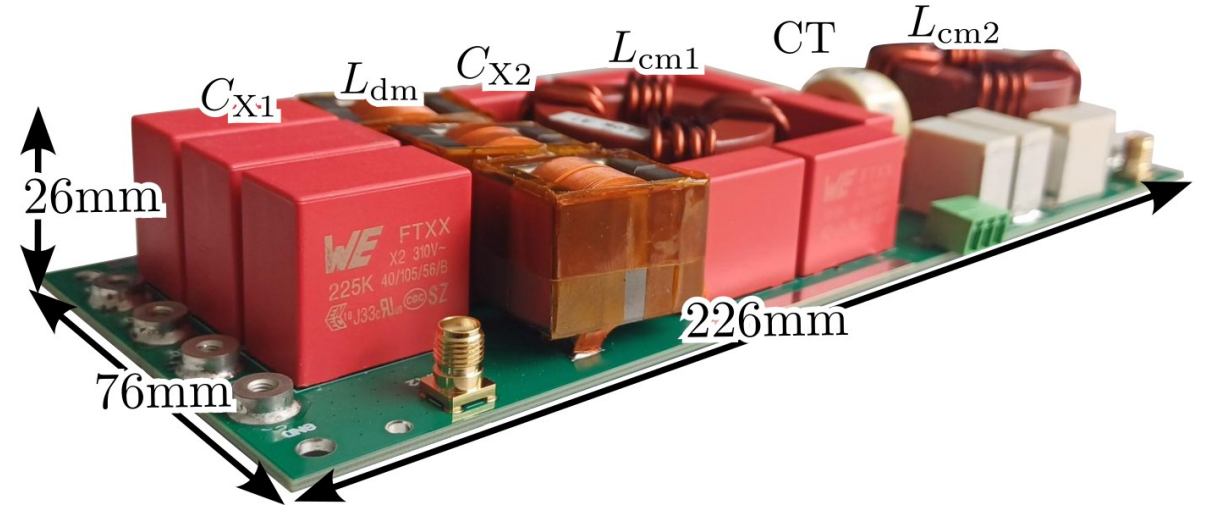
Design Consideration

- Output Capability
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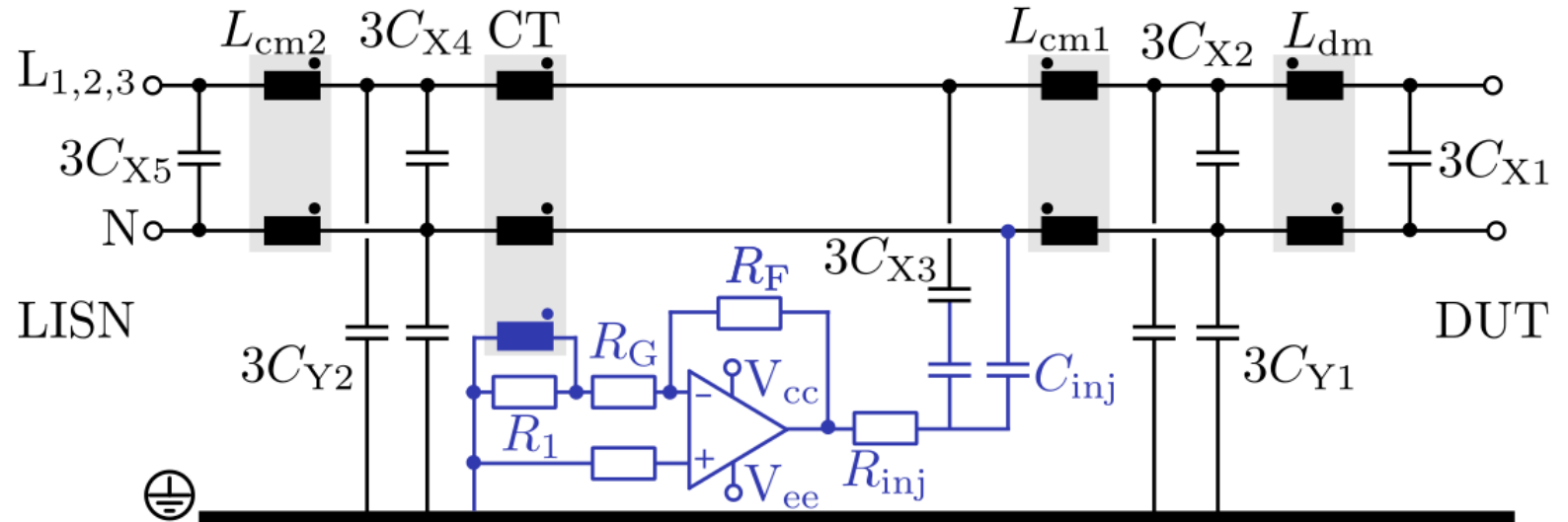
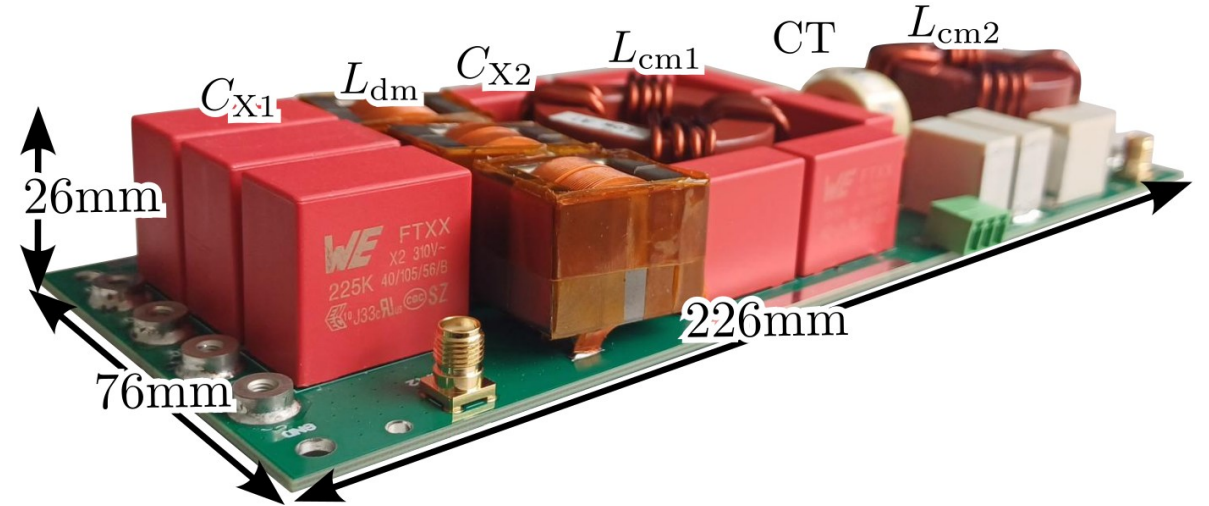
11kW Hybrid Filter (3 ϕ Mode)

- 3 ϕ /1 ϕ Input Compatible
- 1.3mA Leakage Current



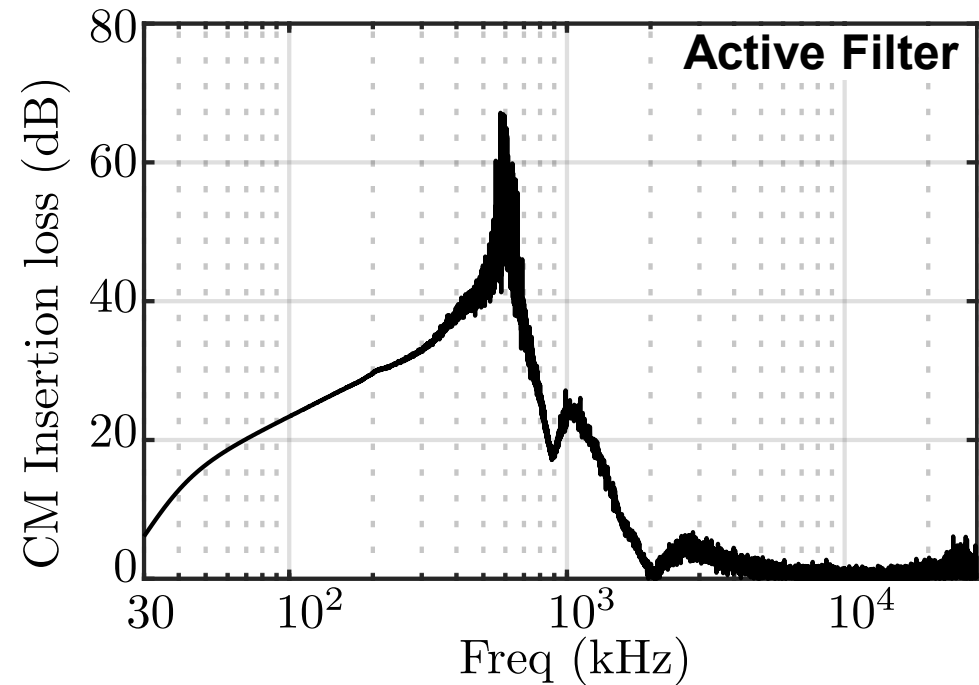
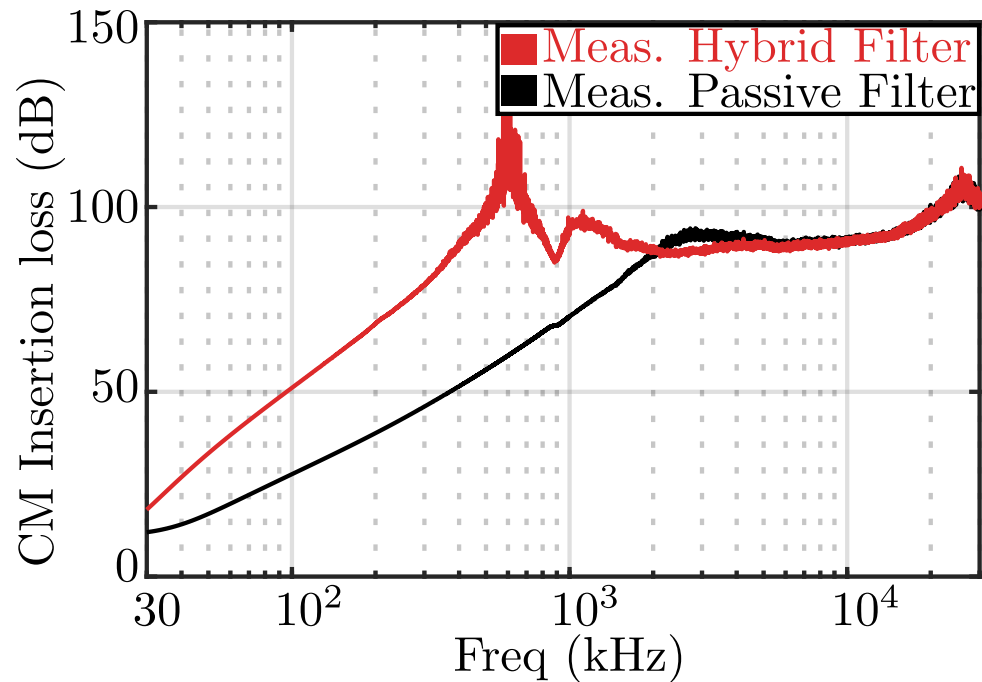
11kW Hybrid Filter (1 ϕ Mode)

- 3 ϕ /1 ϕ Input Compatible
- 1.3mA Leakage Current



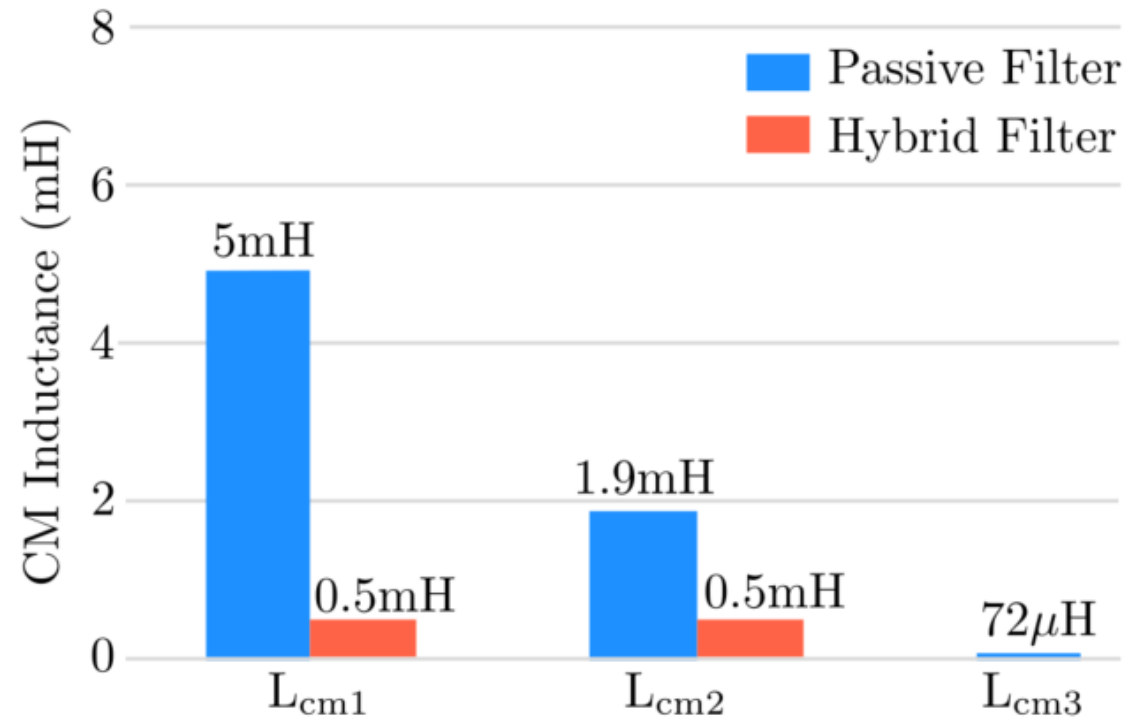
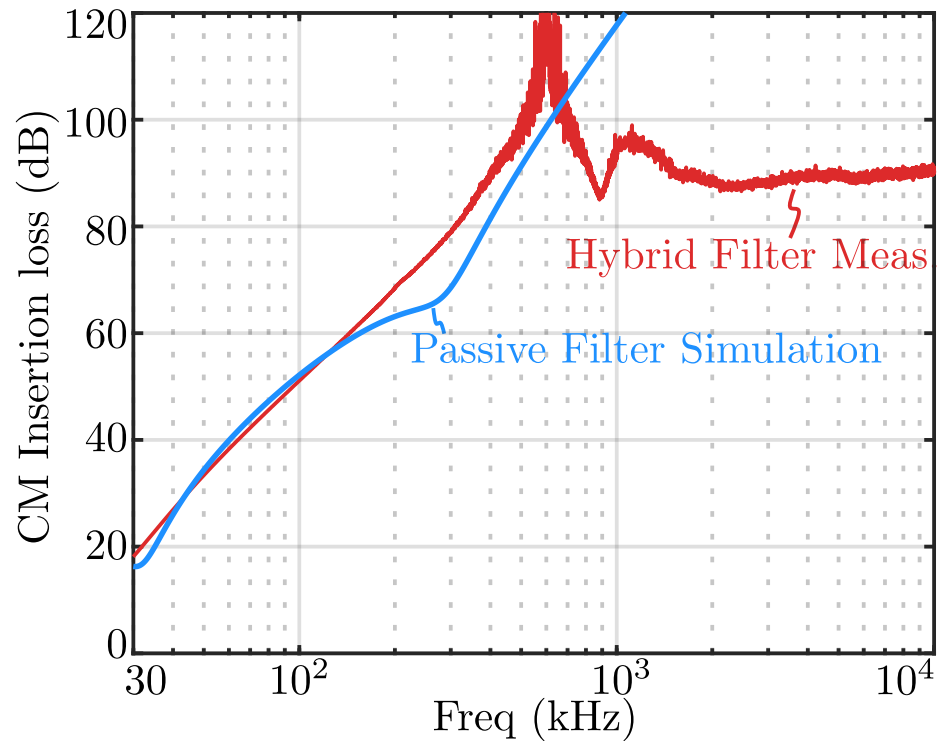
Insertion Loss Measurement

- Effective **30kHz-2MHz**
- **45dB** improvement @450kHz



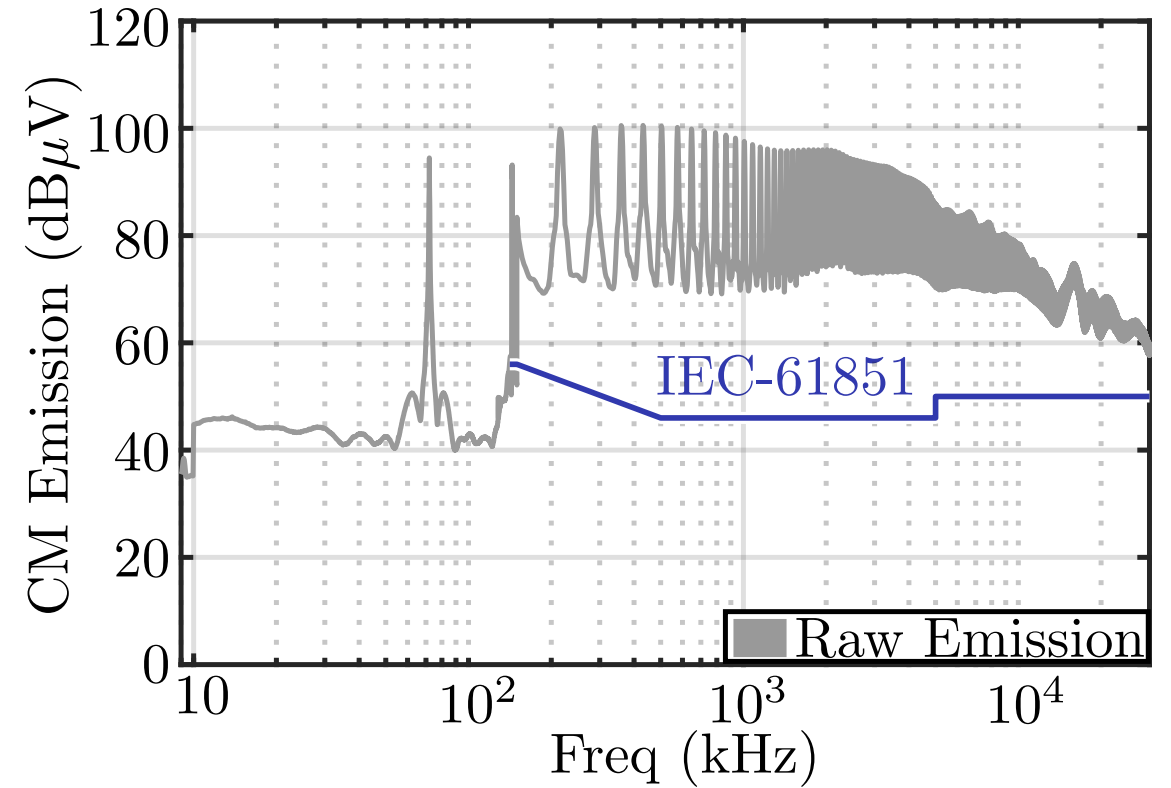
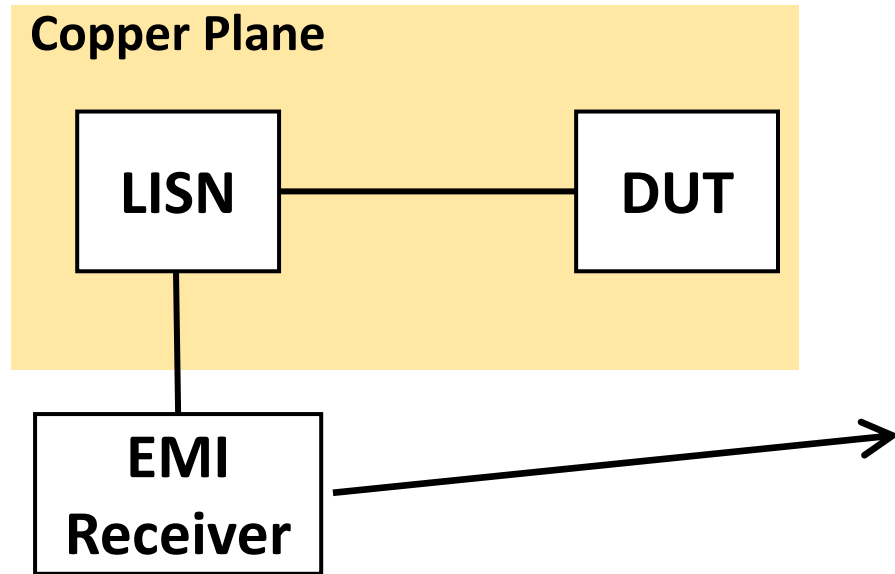
Benchmark Passive Filter

- Lcm1 reduced by **90%**
- Lcm2 reduced by **75%**



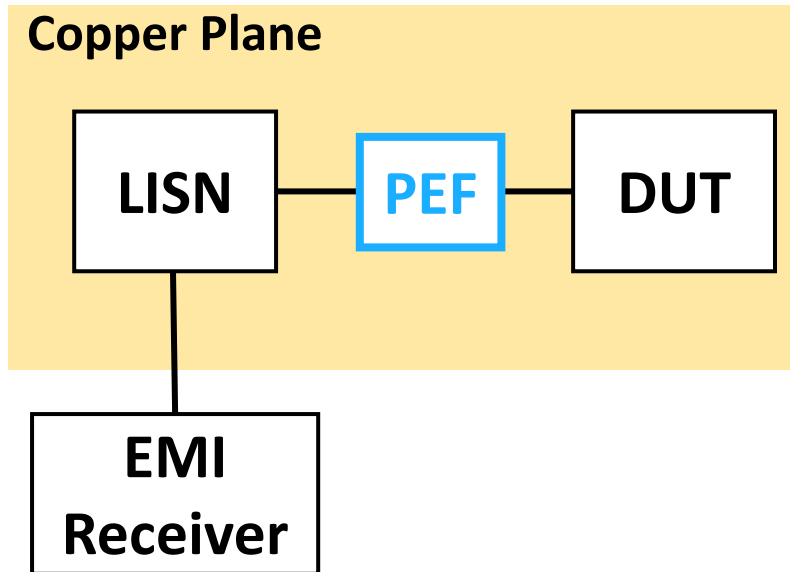
Conductive Emission Measurement

- **50dB** attenuation required

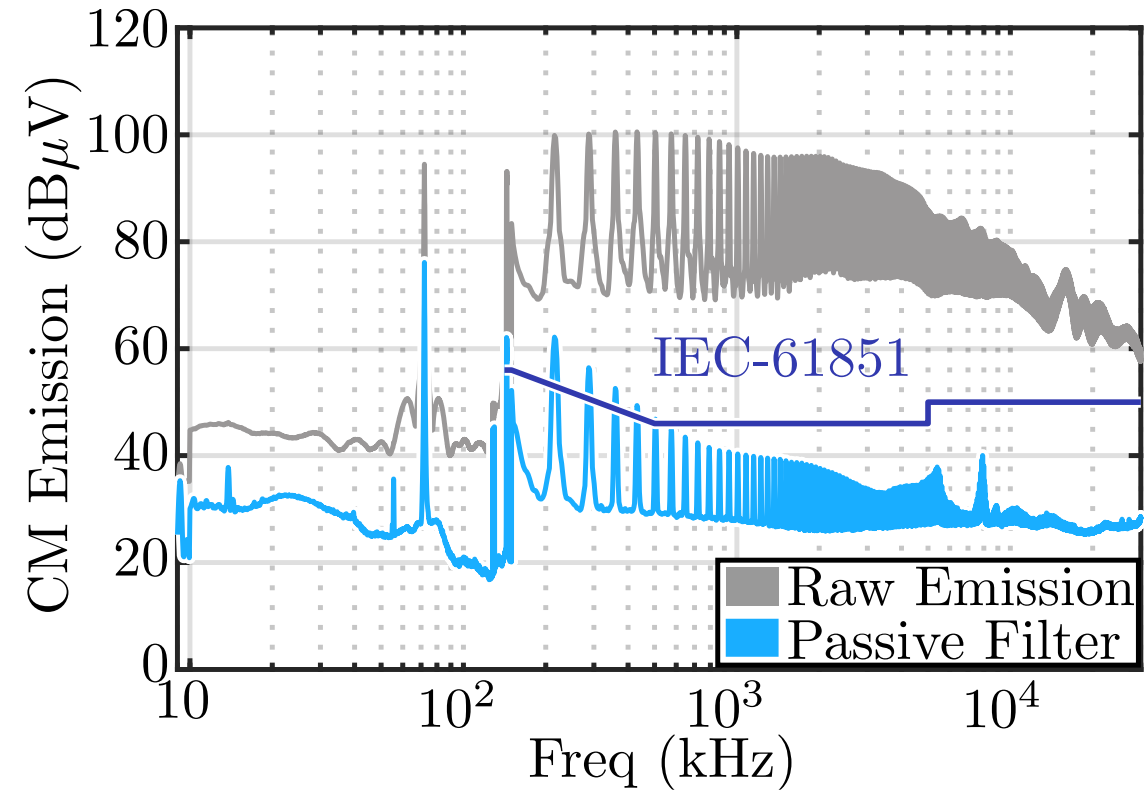


Conductive Emission Measurement

- **10dB** violation in 150kHz-600kHz
- HF Noise is reduced

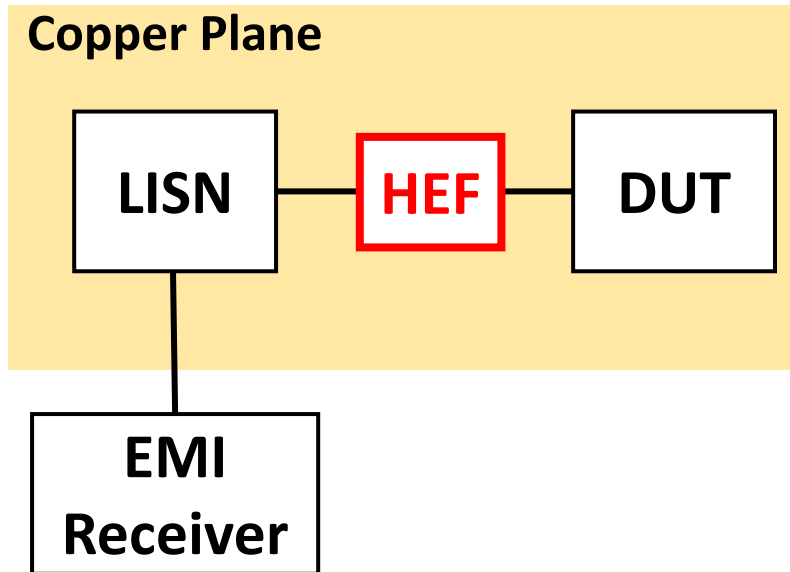


***PEF: Passive EMI Filter**

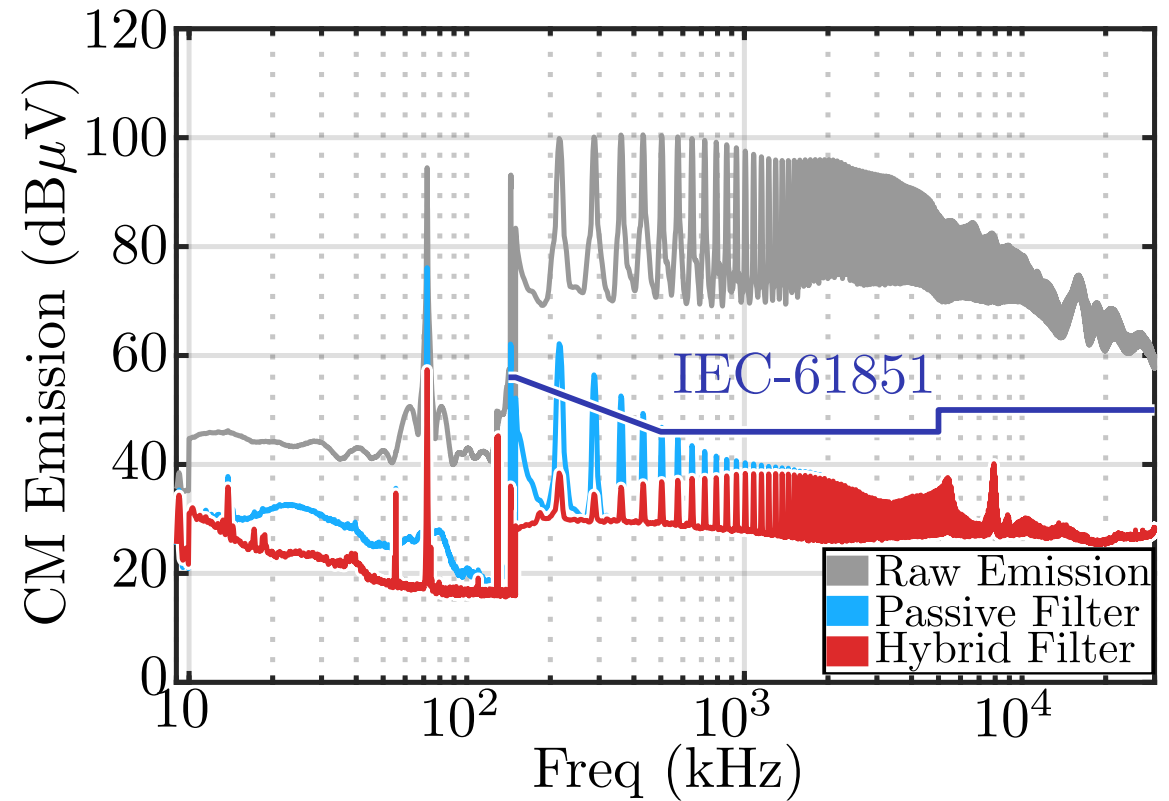


Conductive Emission Measurement

- Effective **9kHz-1MHz**
- **27dB@150kHz**



***HEF: Hybrid EMI Filter**



Conclusion

- Hybrid EMI Filter New Design Option
 - Volume/Weight Reduction
 - Higher Attenuation
 - Higher Power Density

Passive Filter

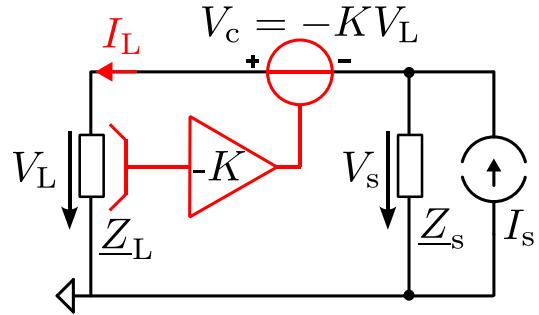


Hybrid Filter



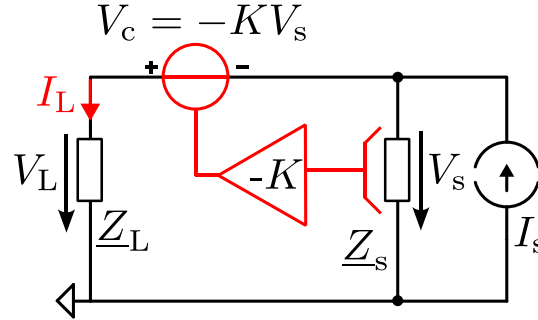
Active EMI Filter Design

Voltage Sense Voltage Compensation



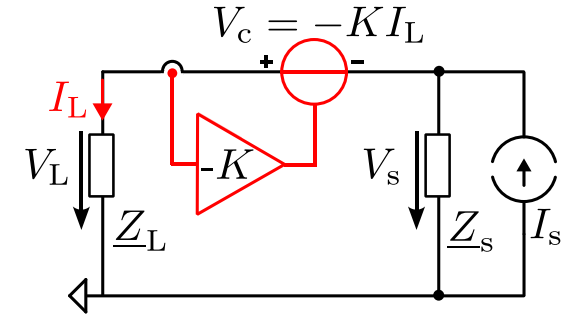
Insertion loss: $1 + \frac{KZ_L}{Z_L + Z_s}$
 Effective range: $|Z_L| \gg |Z_s|$

Voltage Sense Voltage Compensation



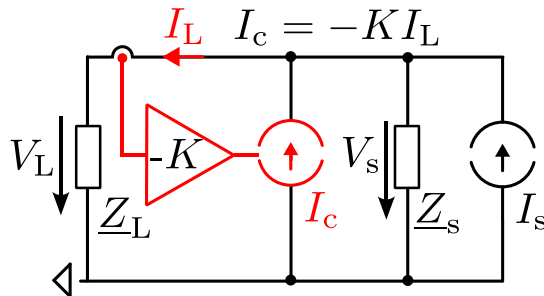
Insertion loss: $\frac{1}{1-K} \left(1 - \frac{KZ_s}{Z_s + Z_L}\right)$
 Effective range: $K = 1, |Z_L| \gg |Z_s|$

Current Sense Voltage Compensation



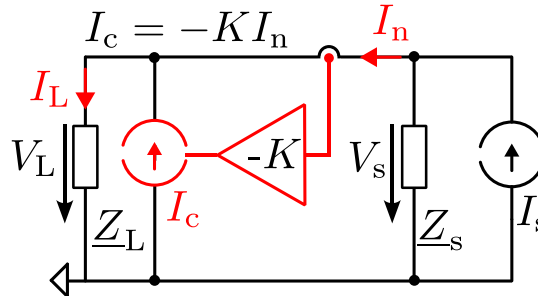
Insertion loss: $1 + \frac{K}{Z_s + Z_L}$
 Effective range: $K \gg |Z_L + Z_s|$

Current Sense Current Compensation



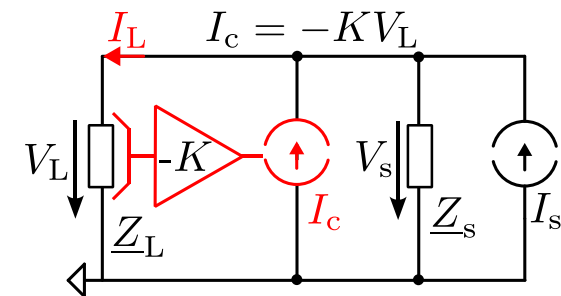
Insertion loss: $1 + \frac{KZ_s}{Z_L + Z_s}$
 Effective range: $|Z_s| \gg |Z_L|$

Current Sense Current Compensation



Insertion loss: $\frac{1}{1-K} \left(1 - \frac{KZ_L}{Z_s + Z_L}\right)$
 Effective range: $K = 1, |Z_s| \gg |Z_L|$

Voltage Sense Current Compensation



Insertion loss: $1 + \frac{KZ_s Z_L}{Z_s + Z_L}$
 Effective range: $K \gg \left|\frac{1}{Z_L} + \frac{1}{Z_s}\right|$

Conclusion

3 ϕ / 1 ϕ Comptible 11kW Hybrid EMI Filter

- Inductance reduced by **90%**
- **27dB** attenuation improvement @**150kHz**
- Effective from **9kHz-1MHz**
- **Low-Profile** Filter Design (26mm)

Passive Filter



Hybrid Filter



Insertion Loss Measurement

